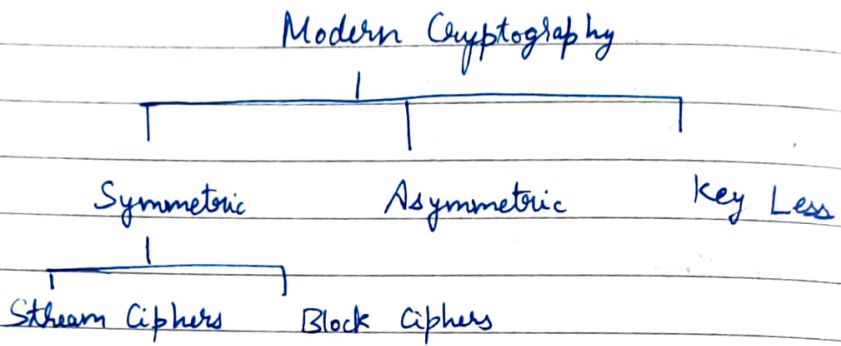


18th September, 2024

UNIT 2

L1



→ SKC

Properties:

1. Confidentiality
2. Integrity
3. Authenticity
4. Data encryption & decryption
5. Secure communication
6. File & Disk encryption

SKC encryption & decryption is done quicker with less processing power.

Advantages of SKC:

1. Speed and efficiency
2. Scalability
3. Simplicity

→ Stream Ciphers

- * bit by bit to transform plain text into code that's unreadable without key
- * linear → same key for encrypt & decrypt
- * Aren't safe

Disadvantages of SKC:

- Key management
- Key distribution
- Key compromise
- Key exhaustion
- Authentication
- Non-repudiation



* Time varying transformations

* uses keystreams to generate ciphertext

* cipher key (128/256 bits) & a nonce digit (64-128 bits)
to produce keystream - pseudorandom number
XORed with plain to produce cipher.

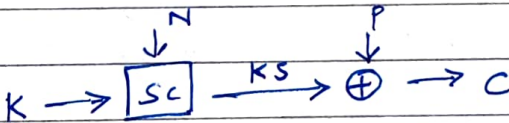
* key, nonce $\rightarrow$ can be reused
keystream $\rightarrow$ has to be unique

* feedback shift registers - unique nonce
(no used only once)

$$KS = SC(K, N)$$

encrypt $C = P \oplus KS_C = P$

decrypt $P = C \oplus KS_P = C$



$\rightarrow$ Stream ciphers: Making OTP practical

cannot have perfect security

Def: PRG is unpredictable if it is not predictable
 $\forall i$: no "eff" adv. can predict bit $(i+1)$
for non-negative E

we say

$$G: K \rightarrow \{0,1\}^n \text{ is predictable if } \Pr[A(G(K))_1 \dots_i = G(K)_{1+i}] \geq \frac{1}{2} + \epsilon$$

eg: Rivest cipher (RC4)

Salsa 20

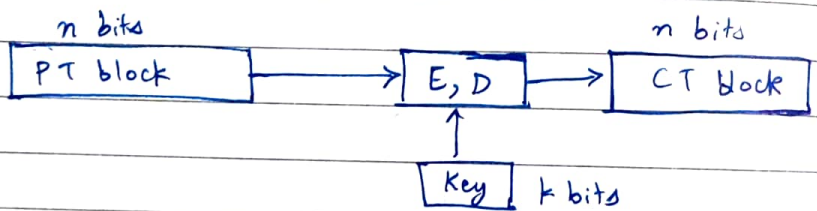
PANAMA, Scream, Rabbit, HC-256 & Grain

- Benefits of Stream Cipher :
1. Speed
 2. Low complexity
 3. Serial nature
 4. Ease of use.

* Best way to ensure security in SC :

- i) use keys just once
- ii) key should be unrelated to any other used.

→ Block Cipher



canonical egs:

1. 3DES : $n = 64$ bits , $k = 168$ bits
2. AES : $n = 128$ bits , $k = 128, 192, 256$ bits

* pseudorandom permutation (PRP) families that operate on fixed size block of bits

Adv : * Block cipher has high diffusion

* Slower than Stream cipher

Characteristics :

- * Confusion - Substitution
- * Diffusion - Transposition / Permutation

Ciphers with adequate diffusion satisfy - avalanche criteria of cryptography

→ Block size consideration

1. Avoid small block size - dictionary attack
2. Avoid blocks of bigger size - difficult to operate with
3. 8-bit multiples - easy to process, don't need padding

egs: DES, Triple DES, AES, Serpent, Twofish

Design Principle

- i) Block size $\uparrow$ security $\uparrow \Rightarrow$ slows cipher
- ii) Key size $\uparrow$ security $\uparrow \Rightarrow$ slows cipher
- iii) no. of rounds $\uparrow$ " "
- iv) subkey generation $\uparrow$ " "
- v) Round function $\uparrow$ " "

L2 Random numbers :
unpredictable initial randomness (entropy)

- Properties:
1. Unbiased (Uniform Distribution)
 2. Unpredictable (independence)
 3. Unreproducible

→ Pseudo: Random Number Generation (PRNG)
|
use initial seed value

→ True Random Number Generation (TRNG)
|
use physical processes to generate randomness
electronic noise, radioactive decay, atmospheric noise
higher unpredictability than PRNG
more suitable for cryptography

Pseudorandomness : Finding a ~~secret~~ deterministic (polynomial-time) algo

$f: \{0, 1\}^n \rightarrow \{0, 1\}^{l(n)}$ such that:

OP of $f(x)$ contains $l(n)$ bits

$\{0, 1\}^{l(n)}$ appears to be as "random looking" as a truly random string

Applications:

1. Key generation
2. nonces
3. salts in certain sign schemes
4. Generation of Primes

L3 Feistel cipher : used by DES

Shannon structure $\begin{cases} \text{Confusion} \\ \text{Diffusion} \end{cases}$

* Substitution Permutation (S-P) networks
modern substitution-transposition product cipher

SP-network — Substitution (S-box)
Permutation (P-box)

— Diffusion : making each Plaintext bit affect as many ciphertext bits as possible
1 bit change in PT, significant effect on CT

eg: Transposition & Permutation.

- Confusion : Making relationship b/w CT & key as complex as possible.

Relationship b/w CT & key is obscured

eg: Substitution

Product Ciphers

- 1st : * invertible & non invertible components

- * Feistel ciphers : DES

- 2nd : * invertible components

- * non-Feistel ciphers : AES

→ Feistel Cipher : Invertible product cipher

Partition IP 2 halves

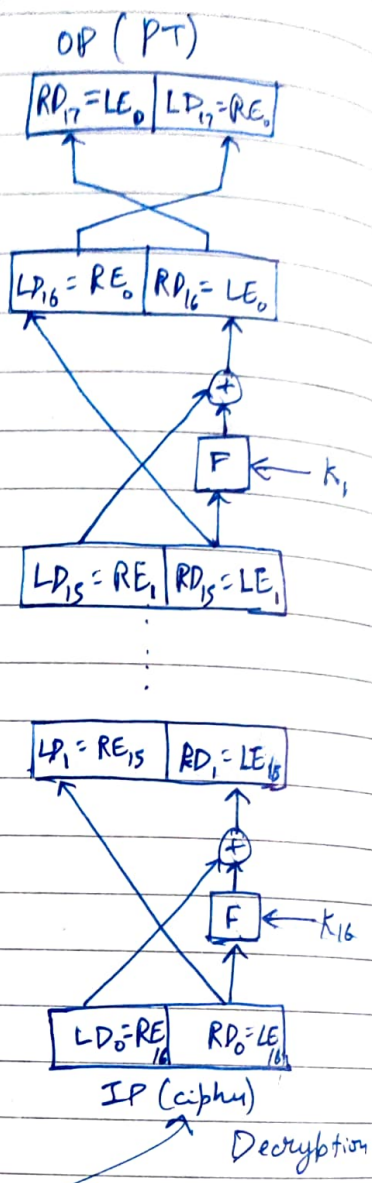
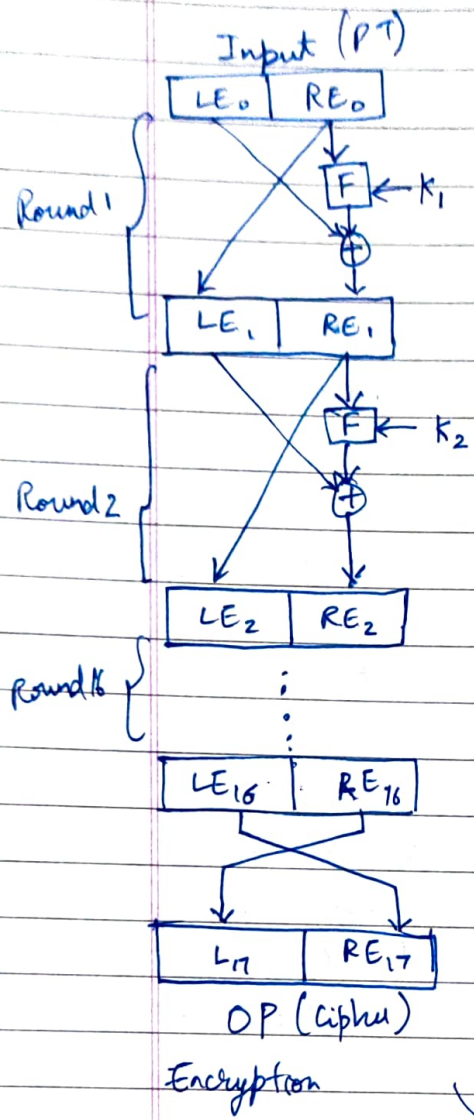
- Process through multiple rounds
- Substitution on Left half
- Round func on right half
- then have permutation swapping halves

Decryption : 1. IP to decryption algo is CT
Rules block produced by encryption algo

2. The seq. of subkeys used in encryption are reversed.

Design : 1. Block Size
Features 2. Key Size
3. No. of rounds
4. Subkey generation algo
5. Round func.
6. Ease of analysis
7. Fast SW encryption/decryption

NESO V1 Feistel Structure (Block Cipher)



NESO V2 DES : Symmetric Block Cipher
 (Data Encryption Standard) (group of bytes IP)
 outdated

easily hackable, replaced by AES

- | | |
|--------------|---------------------|
| IP : 64 bits | Main Key : 64 bits |
| OP : 64 bits | Sub Key : 56 bits (|
| | RoundKey : 48 bits |
| | no. of rounds : 16 |

64 bit Plain Text

64 bit key

Initial Permutation

Permuted choice 1

64

56

Round 1

K_1 48

Permuted choice 2

56

Left circular Shift

64

56

Round 2

K_2 48

Permuted choice 2

56

Left circular Shift

...

...

Round 16

K_{16} 48

Permuted choice 2

56

Left circular Shift

32 bit swap

64

Inverse Initial Permutation

64 bit Cipher Text

$LH_{i-1}(32)$

$RH_{i-1}(32)$

Mangled key

Expansion Permutation

48

$\oplus$

48

Keyed Substitution

32

Transposition

$\oplus$

RH_{i-1}

32

RH_i

LH_i

key

LH_{i-1}

RH_{i-1}

32

Left Shift

32

Left Shift

Contraction Permutation

PC 2

48

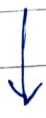
Round Key_i

LHK_{i-1}

RHK_{i-1}

→ Initial Permutation

1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24
25	26	27	28	29	30	31	32
33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48
49	50	51	52	53	54	55	56
57	58	59	60	61	62	63	64

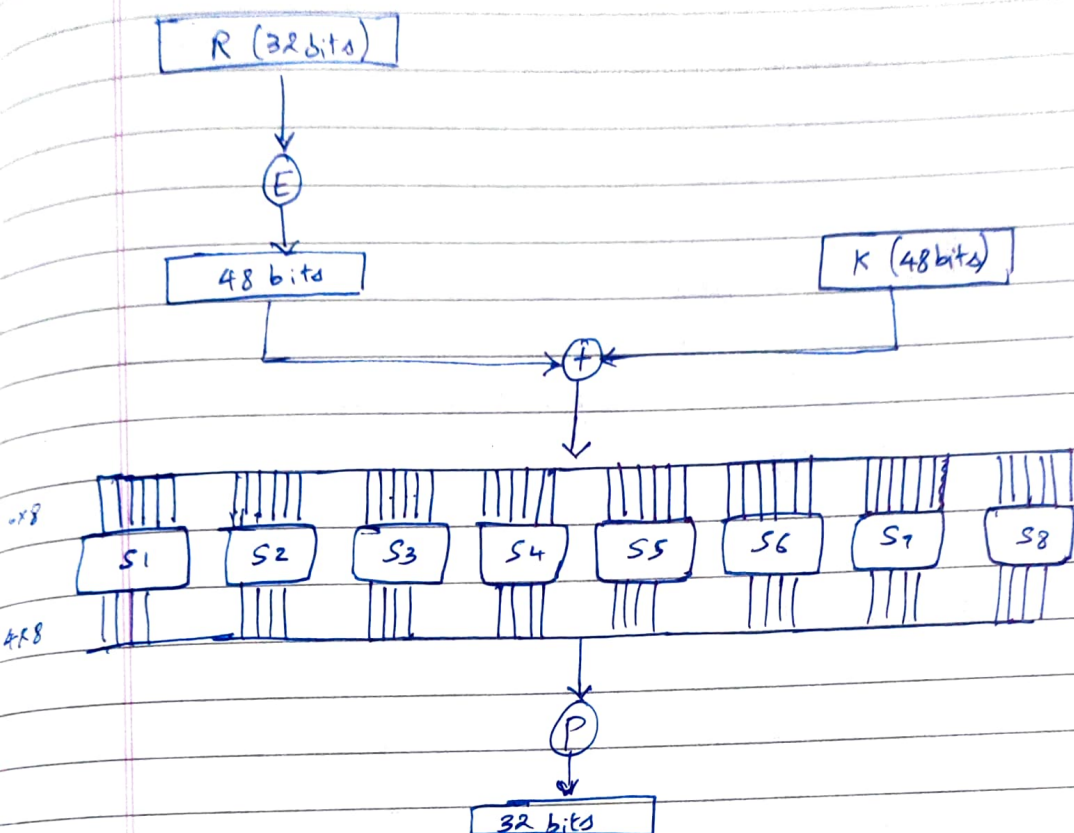


58	50	42	34	26	18	10	<u>2</u>
60	52	44	36	28	20	12	<u>4</u>
62	54	46	38	30	22	14	<u>6</u>
64	56	48	40	32	24	16	<u>8</u>
57	49	41	33	25	17	9	<u>1</u>
59	51	43	35	27	19	11	<u>3</u>
61	53	45	37	29	21	13	<u>5</u>
63	55	47	39	31	23	15	<u>7</u>

→ Inverse Initial Permutation

40	8	48	16	56	24	64	32
39	7	47	15	55	23	63	31
38	6	46	14	54	22	62	30
37	5	45	13	53	21	61	29
36	4	44	12	52	20	60	28
35	3	43	11	51	19	59	27
34	2	42	10	50	18	58	26
<u>33</u>	<u>1</u>	<u>41</u>	<u>9</u>	<u>49</u>	<u>17</u>	<u>57</u>	<u>25</u>
	<u>I</u>		<u>II</u>		<u>III</u>		<u>IV</u>

Mangler Func.



⑤ Expansion Matrix Permutation

32	1	2	3	4	5
4	5	6	7	8	9
8	9	10	11	12	13
12	13	14	15	16	17
16	17	18	19	20	21
20	21	22	23	24	25
24	25	26	27	28	29
28	29	30	31	32	1

Strength of DES

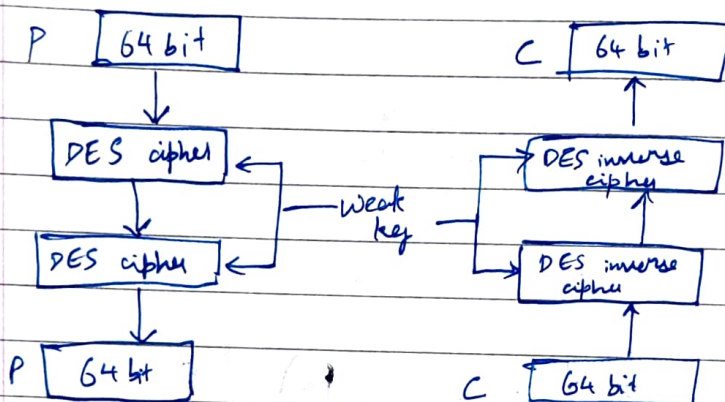
Criticisms

- Reduction in key size
- S-boxes were classified

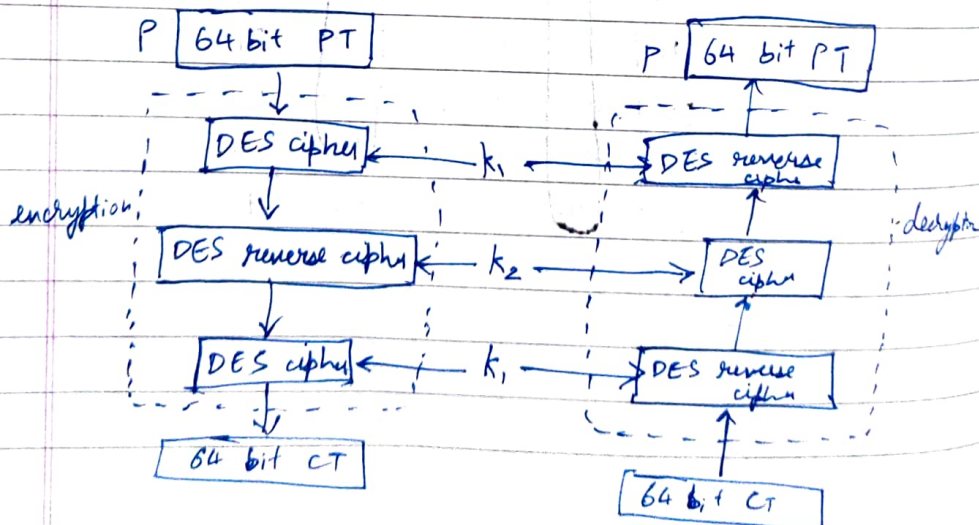
Avalanche effect in DES (strong)

If small change in either the Plain Text or the key, the ciphertext should change markedly.

→ Double DES



→ Triple DES (with 2 key)



Groups
Abelian groups

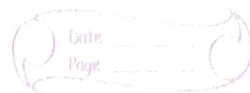
Rings

Commutative Rings

Integral domains

Fields

Finite fields



L4

→ Groups:

A group (G) is a set of elements with a binary operation $(\cdot)$ that satisfies 4 properties

① Closure: $a, b \in G, c = a \cdot b \in G$

② Associativity: $a, b, c \in G$
 $(a \cdot b) \cdot c = a \cdot (b \cdot c)$

③ Identity: $e \in G$
 $\forall a \in G$
 $a \cdot e = e \cdot a = a$

④ Inverse

$\forall a \in G \exists a^{-1} \in G$
 $a \cdot a^{-1} = a^{-1} \cdot a = e$

Abelian group: Above props + Commutativity

$a, b \in G$
 $a \cdot b = b \cdot a$

→ Cyclic Group

if every element $b \in G$ is power of some element a
 $b = a^k$

$a \rightarrow$ generator of group G

* cyclic group is always abelian & may be finite/infinite

$\rightarrow$ Rings $\{R, +, \times\}$
satisfy all axioms:

1. Abelian Group under addition ($A_1 \rightarrow A_5$)

2. Closure under multiplication (M_1)

$$a, b \in R_g$$

$$c = ab \in R_g$$

(If $a, b \in R$ then ab also $\in R$)

3. Associativity of multiplication (M_2)

$$a, b, c \in R_g$$

$$(ab)c = a(bc) \quad \text{for all } a, b, c \in R$$

4. Distributive (M_3)

$$a, b, c \in R_g$$

$$a(b+c) = ab+ac$$

$$(a+b)c = ac+bc \quad \text{for all } a, b, c \in R$$

5. Commutative (M_4)

$$ab = ba \quad \text{for all } a, b \in R$$

integral domain $\Rightarrow$ Multiplicative identity (M_5)

$$a1 = 1a = a \quad \text{for all } a \in R$$

No Zero Divisors

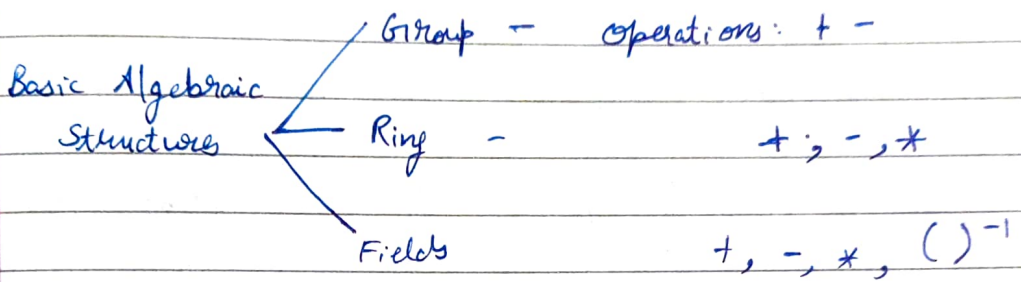
$$a, b \in R \quad \text{if } ab = 0$$

then either $a=0$ or $b=0$

Field : everything else + Multiplicative Inverse ($\neq 0$)

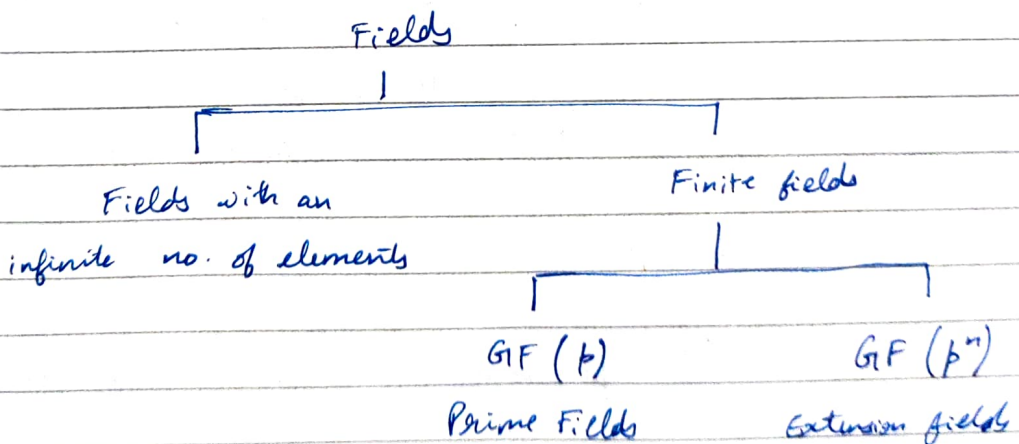
$$a \in S \text{ et } a \neq 0$$

$$a^{-1} \text{ in } S \text{ such that } aa^{-1} = a^{-1}a = 1$$



Galois Fields (Finite field)

order of finite field (no. of elements in field)
must be p^m elements, $p \rightarrow$ prime
 $m \rightarrow$ anything



$\rightarrow$ Prime Fields GF(p) $0, 1, \dots, p-1$
GF(5) = $\{0, 1, 2, 3, 4\}$

- Add, Sub, Mul

$a, b \in GF(p) = \{0, 1, \dots, p-1\}$ then

$$a + b \equiv c \pmod{p}$$

$$a - b \equiv c \pmod{p}$$

$$a \cdot c \equiv c \pmod{p}$$

Inversion

$$a \in GF(p)$$

$$a \cdot a^{-1} = 1 \pmod{p}$$

→ Extension Fields - polynomials

Extension fields = $GF(2^m)$ where $m > 1$

$$GF(2^3) = GF(8)$$

$$(a_2, a_1, a_0)$$

$$(0, 0, 0) = 0$$

$$(0, 0, 1) = 1$$

$$(0, 1, 0) = x$$

$$(1, 0, 0) = x^2$$

$$(0, 1, 1) = x + 1$$

$$(1, 1, 0) = x^2 + x$$

$$(1, 0, 1) = x^2 + 1$$

$$(1, 1, 1) = x^2 + x + 1$$

$$GF(2^3) = \{0, 1, x, \dots\}$$

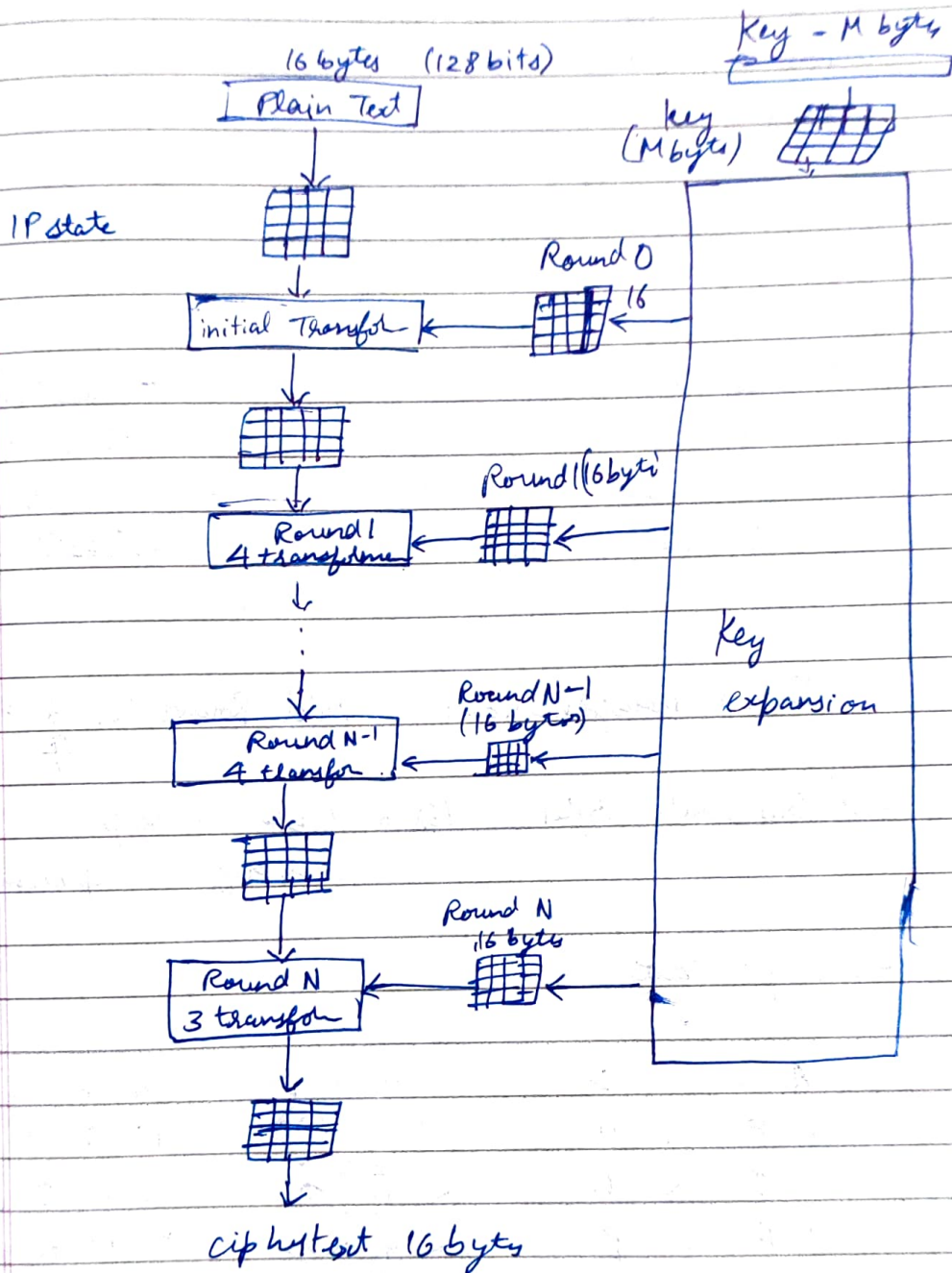
→ Irreducible polynomials, prime polynomials

cannot be factored into further over rational numbers.

- Security
- Performance
- Efficiency
- Ease of Impl
- Flexibility

AES Advanced Encrypted Standard

* Symmetric Block cipher



no. of rounds	Key size (in bits)
10	128
12	192
14	256

AES 128

AES 192

AES 256

Key Size	128	192	256
Plain Text Size	128	128	128
No. of Rounds	10	12	14
Round Key Size	128	128	128

AES Transformation funcns

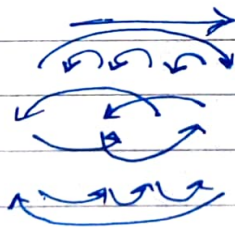
- ① Substitute Bytes - Substitution
- ② Shift Rows - Permutation (Diffusion)
- ③ Mix Columns (not in last Round) - Substitution (Diffusion)
- ④ Add Round Key - Substitution

Initial Transformation: Add Round Key to PT $\oplus$

- ① $\rightarrow$ Substitute Bytes 16x16 table lookup
(Mapping $\rightarrow$ replacement)

- ② $\rightarrow$ Shift Rows

A	B	C	D
E	F	G	H
I	J	K	L
M	N	O	P



A	B	C	D
F	G	H	E
K	L	I	J
P	M	N	O

- ③ $\rightarrow$ Mix Columns : Matrix multiplication

- ④ $\rightarrow$ Add Round Key : XOR operation

L # Modes of Operation (5)

1. Electronic Codebook (ECB)
2. Cipher Block Chaining (CBC)
3. Cipher Feedback (CFB)
4. Output Feedback (OFB)
5. Counter (CTR)

① → Electronic Codebook

- * Same PT $\Rightarrow$ Same CT
- * Each block encoded independently using same key.

Pros:

- i) Simplest mode
- ii) Ideal for small data
- iii) Independent
- iv) Fast - Parallelism

Cons:

- i) Not secure for lengthy messages
- ii) Cryptanalyst can exploit the regularities of ^{the} msg

② → Cipher Block Chaining (CBC)

- * Same PT block $\Rightarrow$ Different CT block
- * same key
- * chaining
- * dependent block
- * $\therefore$ repeating patterns of bits are not exposed
- * general purpose block oriented transmission

Pros: i) Appropriate mode for encrypting messages of length $> 'b'$ bits

ii) Confidentiality + Authentication

Cons: i) Slow - No Parallelism

ii) cannot encrypt any block since we need the cipher text of previous block.

iii) Initialization Vector (IV) which must be known to sender & receiver.

③ → Cipher Feedback (CFB)

* Block cipher → Stream cipher

└ No need of padding

└ Real Time

└ len of PT = len of CT

* general purpose stream oriented transmission

* Authentication

Pros: i) can operate in real time

ii) need of padding is eliminated

iii) Encryption func does decryption as well

iv) len of PT = len of CT

Cons: i) Chances of wastage of transmission capacity

ii) Not a typical stream cipher

④ → Output Feedback (OFB)

* No selection of bits, all bits used.

- Pros:
- i) Bit errors in transmission do not propagate
 - ii) Same PT - Same key - Different CT
 - iii) PT len can be random choice

- Cons:
- i) Sender & Receiver must
 - ii) More vulnerable to modification attack
 - iii) No parallelizable
 - iv) IV and keys must be regenerated every time

⑤ → Counter Mode (CTR)

Application : ATM, IP Sec

Size of counter = plaintext block size
diff counter value for each plaintext
Counter value is initialized
counter++

Decryption — Same seq. of counter values used
Initial value of counter is made avail.

- Pros:
- i) Hardware efficiency
 - ii) Software efficiency
 - iii) Pre processing
 - iv) Random access
 - v) Probable security
 - vi) Simple

Can 128bit AES be broken

- * AES itself is unbreakable when implemented properly but it is not always
- * Side-channel attacks

Random Oracle Padding Attack

Vulnerability

'Oracle' - system that performs cryptographic actions by taking IP

Allows attacker to retrieve the original PT from CT w/o knowing the key

- 4 main things for padding attack:
- i) XOR operation
 - ii) CBC Mode
 - iii) Padding Standard
 - iv) Oracle

→ XOR operator

changing 1 byte will only affect the byte its ^{byte} ~~lane~~ ^{done by} ~~lane~~ ^{byte} in

→ Padding Oracle

* Application decrypts encrypted data; first decrypt data
↓
remove padding

* During cleanup of padding, if an invalid padding triggers a detectable behaviour → Padding Oracle Vulnerability

error, lack of results, slower response.

→ Security Advantages of AES

1. Resistance to known attacks
2. Adaptability
3. Widespread adoption

* Force the decrypted PT to have valid Padding by manipulating the IP CT

Need for Asymmetric Key Cryptography

- Secure communication
- Identity verification
- Digital signatures
- PKI
- Web security
- Email security

PT: Cryptanalysis

a b c d e f g h i j k l m n o p q r s t u v w x
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23

A-10

C R Y P T A N A L Y S I S

B-11

2 17 24 15 19 0 13 0 11 24 18 8 18

C-12

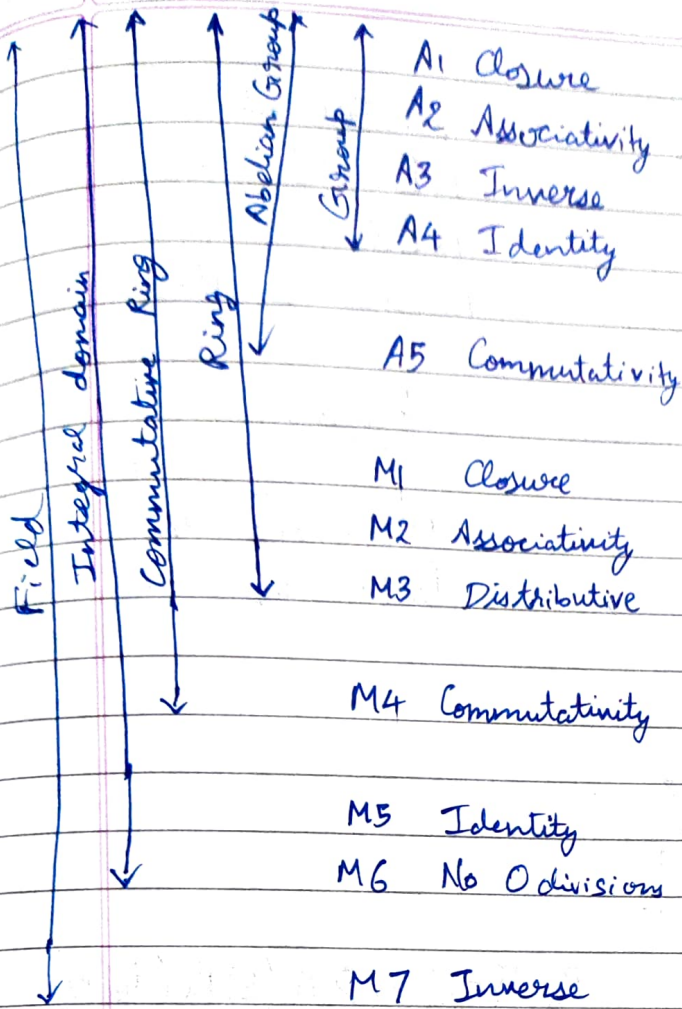
D-13

hex: 2 11 18 0F 13 00D 0 B 18 12 8 12

E-14

F-15

62	13	08	82
11	05	18	19
18	0D	12	19
0F	0	18	19



DES	AES
* Bit Oriented	Byte oriented
* Feistel NW	Substa Permutation NW
* Rounds = 16	Rounds depend on Key size
4 2 24 25	

Plain Text set

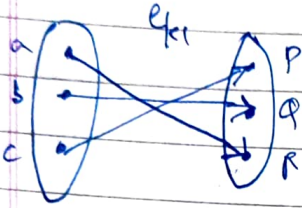


Plain Text set

$$P_X[X=a] = 1/2$$

$$P_X[X=b] = 1/3$$

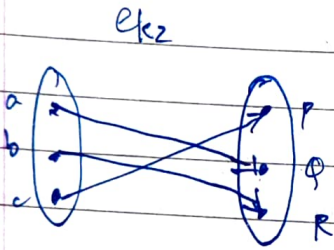
$$P_X[X=c] = 1/6$$



keyspace

$$P_K[K=k_1] = 3/4$$

$$P_K[K=k_2] = 1/4$$



$$P_Y[Y=y] = \sum_k P_K(k) P_X(d_k(y))$$

$$\begin{aligned} P_Y[Y=P] &= P_K(k_1) P_X(c) + P_K(k_2) P_X(c) \\ &= \frac{3}{4} \left(\frac{1}{6} \right) + \frac{1}{4} \left(\frac{1}{6} \right) = \frac{1}{6} \end{aligned}$$

$$P_Y[Y=Q] = \frac{1}{3} \left(\frac{3}{4} \right) + \frac{1}{2} \left(\frac{1}{4} \right) = \frac{3}{8}$$

$$P_Y[Y=R] = \frac{1}{2} \left(\frac{3}{4} \right) + \frac{1}{3} \left(\frac{1}{4} \right) = \frac{11}{24}$$

Bayes Theorem

$$\text{Posteriori Prob} = P[X=x|Y=y]$$

$$P[x|y] = \frac{P[x] \times P[y|x]}{P[y]}$$

$$P[y|x] = \sum P_k[k]$$

9.

a b c d e f g h i j k l m n o p q r s t u v w x y z
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25

P R O S P E R

15 17 14 18 15 4 17

B O P B O P B

1 14 15 1 14 15 1

16 31 30 19 29 19 18 mod 26

16 5 3 19 3 19 18

Q F D T D T S

$$28 \div 7 = 4$$

ANALYST

A	A	L	N	S	T	Y
T	E	C	H	N	T	E
R	L	I	A	T	E	N
L	I	G	L	N	C	E
E	G	E	A	C	Y	N

A	N	A	L	Y	S	T
T	H	E	C	E	N	T
R	A	L	I	N	T	E
L	L	I	G	E	N	C
E	A	G	E	N	C	Y

$$(12 - 43) \bmod 13$$

$$(-31) \bmod 13$$

$$8 \bmod 13 = 8$$

$$10^n \bmod x = (10 \bmod x)^n \bmod x$$

$$10^2 \bmod \cdot$$

q	r_1	r_2	r
1	2740	1760	980
1	1760	980	780
1	980	780	200
3	780	200	180
1	200	180	20
9	180	20	0
	20	0	

$$\begin{array}{r}
 17 \\
 20 \overline{) 780} \\
 \underline{60} \\
 180 \\
 3 \\
 200 \overline{) 780} \\
 \underline{600} \\
 180 \\
 + \\
 180 \overline{) 200} \\
 \underline{180} \\
 20
 \end{array}$$

$$S = S_1 - S_2 \times q \quad | \quad t = t_1 - t_2 \times q$$

→ Extended Euclidean Algo
(161, 28)

$$\begin{array}{l}
 S = 1, S_2 = 0 \\
 t_1 = 0, t_2 = 1
 \end{array}$$

q	r_1	r_2	r	S_1	S_2	t_1	t_2	t
5	161	28	21	1	0	1	0	-5
	28	21	7	0	1	0	1	

$$q = r_1$$

$$1 - (-5(1))$$

$$\begin{array}{r} 28 \overline{) 161} \\ \underline{140} \\ 21 \end{array}$$

$$\begin{array}{r} 1 \\ 21 \overline{) 28} \\ \underline{21} \\ 7 \end{array}$$

→ Extended Euclidean

$$s = s_1 - s_2 \times q$$

$$t = t_1 - t_2 \times q$$

$$s_1 = 1 \quad s_2 = 0$$

$$t_1 = 0 \quad t_2 = 1$$

161, 28

q	r_1	r_2	r	s_1	s_2	s	t_1	t_2	t
5	161	28	21	1	0		0	1	-5
1	28	21	7	0	1	-1	1	-5	6
3	21	7	0	1	-1	4	-5	6	-23
	<u>7</u>	0		<u>-1</u>	1		<u>6</u>	-23	
	gcd			s			t		

MI of 11 in $\mathbb{Z}_{26}$

$$s = s_1 - s_2(q)$$

$$1 - (-1)(3)$$

~~$$5 - 6(2)$$~~

$$t = t_1 - t_2(q)$$

$$= -5 - 6(3)$$

$$=$$

$$1 - (-2)(2)$$

$$1 + 4$$

$$-5 - (-2)(1)$$

q	r_1	r_2	r	s_1	s_2	s	t_1	t_2	t
2	26	11	4	1	0	1	0	1	-2
2	11	4	3	0	1	-2	1	-2	5
1	4	3	1	1	-2	3	-2	5	-7
3	3	1	0	-2	1		5	-7	26
x	<u>1</u>	0					<u>-7</u>	26	
	gcd						MI		

MI

$$1 - (-2)(1) \quad 5 - (-7)(3)$$

→ Fermat Little Theorem

I $a^{p-1} \equiv 1 \pmod{p}$
 $p \rightarrow \text{prime}$ & a is not $\div$ by p

II $a^p \equiv a \pmod{p}$
 $p \rightarrow \text{prime}$

q. $6^{10} \pmod{11}$ / prime
 $p-1 = 10$
 $= 1 \pmod{11} = 1$

q. $3^{12} \pmod{11}$
 $=$

$p-1 = 12 \times$

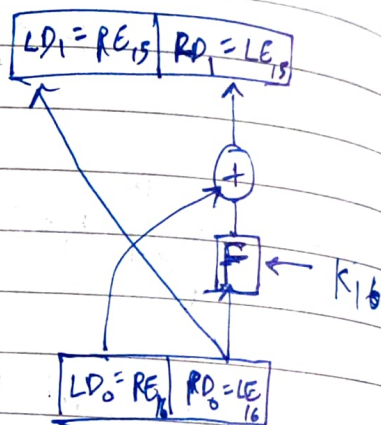
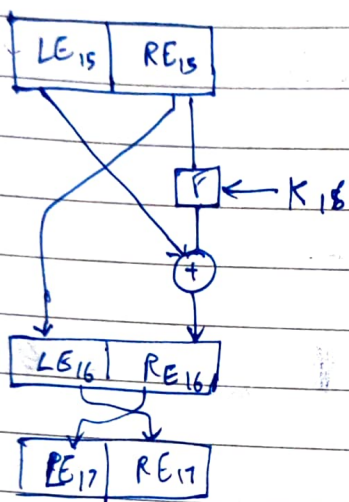
$$\begin{aligned} 3^{12} \pmod{11} &= (3^{11} \times 3) \pmod{11} \\ &= (3^{11} \pmod{11}) (3 \pmod{11}) \\ &= (3 \times 3^{10} \pmod{11}) (3 \pmod{11}) \\ &= 3 \times 1 \times 3 = 9 \end{aligned}$$

→ Fermat & MI

$a^{-1} \pmod{p} = a^{p-2} \pmod{p}$
 $a \in \mathbb{Z}$ & p doesn't $\div$ a

$$\begin{aligned} 8^{-1} \pmod{17} &= 8^{17-2} \pmod{17} \\ &= 8^{15} \pmod{17} \\ &= 15 \pmod{17} \end{aligned}$$

$$\begin{aligned} 5^{-1} \pmod{23} &= 14 \pmod{23} \\ 60^{-1} \pmod{101} &= 32 \pmod{101} \\ 22^{-1} \pmod{211} &= 48 \pmod{211} \end{aligned}$$



$$RE_{16} = LE_{15} \oplus F(K_{16}, RE_{15})$$

Euler's phi func/

Totient func

$$\phi(1) = 0$$

$$\phi(p) = p - 1 \quad (\text{if } p \text{ prime})$$

$$\phi(m \times n) = \phi(m) \times \phi(n) \quad m \& n \rightarrow \text{relatively prime}$$

$$\phi(p^e) = p^e - p^{e-1}$$